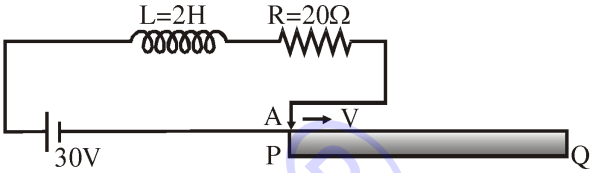
SECTION-I

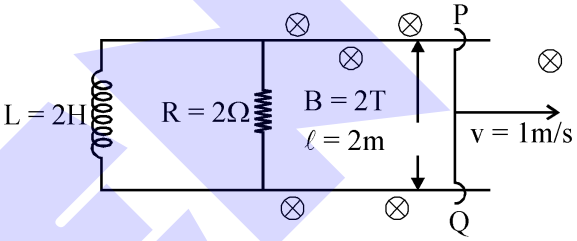
Single Correct Answer Type 8 Q. [3 M (–1)]

1. In the circuit shown in figure, A is a sliding contact which can move over a smooth rod PQ. Resistance per unit length of the rod PQ is 1 ohm/m. Initially slider is just left to the point P and circuit is in the steady state. At t = 0 slider starts moving with constant velocity v = 5 m/s towards right. Current in the circuit at t = 2 sec is :-



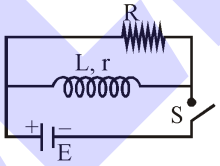
(A) 1 amp
(B) less than 0.5 amp
(C) more than 1 amp
(D) in between 0.5 to 1.0 amp

2. The given figure shows an inductor and resistor fixed on a conducting wire. A movable conducting wire PQ starts moving on the fixed rails from t = 0 with constant velocity 1 m/s. The work done by the external force on the wire PQ in 2 seconds is :-



(A) 16 J
(B) 32 J
(C) 48 J
(D) 64 J

3. A solenoid of inductance L and resistance r is connected in parallel to a resistance R. A battery of emf E and of negligible internal resistance is connected across this parallel combination as shown in figure. The circuit is in steady state. At time t = 0, switch S is opened. Choose the **CORRECT** option(s).



(A) Current in the inductor just after opening of the switch is = $\frac{E(r+R)}{rR}$

(B) Total energy dissipated in the solenoid and the resistor long time after opening of the switch is $\frac{1}{2}L\frac{E^2(R+r)^2}{r^2R^2}$

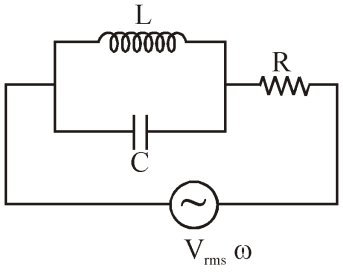
(C) The amount of heat generated in the solenoid after opening of switch is $\frac{E^2L}{2r(r+R)}$

(D) The amount of heat generated in the solenoid after opening of the switch is $\frac{E^2L}{2R(r+R)}$

4. In the given L-C-R circuit the value of rms current through the a.c. source is

(A) $\frac{V_{rms}}{\sqrt{R^2+\left(\omega L-\frac{1}{\omega C}\right)^2}}$

(B) $\frac{V_{rms}}{\sqrt{R^2+\left(\frac{\omega L}{1-\omega^2LC}\right)^2}}$



(C) $\frac{V_{rms}}{\sqrt{R^2+(\omega^2LC-\omega L)^2}}$

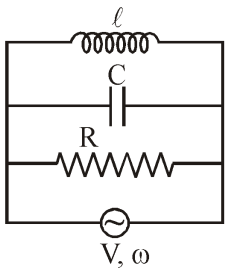
(D) $\frac{V_{rms}}{\sqrt{(R+\omega L)^2+\frac{1}{\omega^2C^2}}}$

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5. Consider L, C, R circuit as shown in figure, with a.c. source of peak value V and angular frequency ω. Then the peak value of current through the ac source

(A) $\dfrac{V}{\sqrt{\dfrac{1}{R^2}+\left(\omega L-\dfrac{1}{\omega C}\right)^2}}$

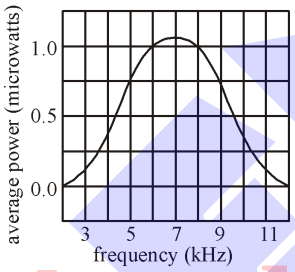
(B) $V\sqrt{\dfrac{1}{R^2}+\left(\dfrac{1}{\omega L}-\omega C\right)^2}$



(C) $\dfrac{V}{\sqrt{R^2+\left(\omega L-\dfrac{1}{\omega C}\right)^2}}$

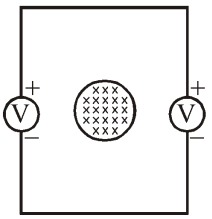
(D) $\dfrac{VR\omega C}{\sqrt{\omega^2C^2+R\left(\omega^2C^2-1\right)^2}}$

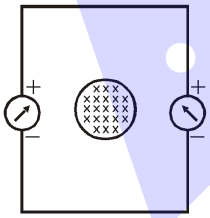
6. The plot given below is of the average power delivered to an LRC circuit versus frequency. The quality factor of the circuit is :

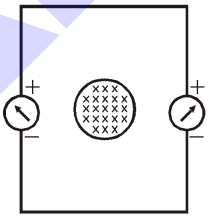


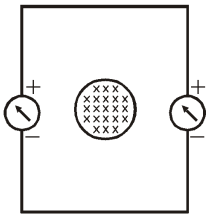
(A) 5 (B) 2.4 (C) 2.8 (D) 1.4

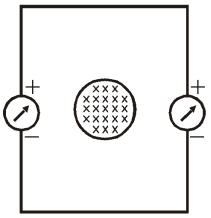
7. Two similar voltmeters with large resistance are connected in a loop as shown. Inside the loop, there is a cylindrical region through which magnetic flux is distributed uniformly. The magnetic field increases uniformly with time. The voltmeters are bidirectional i.e. the needle deflects to right if current in it flows from + terminal to – terminal and needle deflects to left if current in it flows from – terminal to + terminal. Which of the following figures show the readings of voltmeters correctly :-



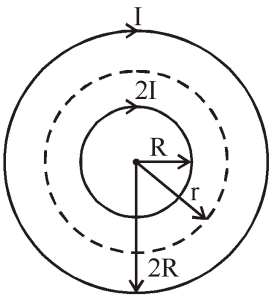
(A) 

(B) 

(C) 

(D) 

8. A long solenoid contains another coaxial solenoid (whose radius R is half of its own). Their coils have same numbers of turns per unit length and initially both carry no current. At any moment the current flowing in the inner coil is twice as large as that in the outer one and their directions are the same. As a result of the increasing current a charged particle initially at rest between the solenoids, starts moving along a circular trajectory. The radius of the circular trajectory is :

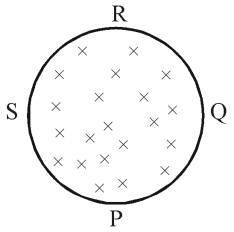


(A) $\dfrac{5}{4}R$ (B) $\sqrt{2}\,R$ (C) $\dfrac{7}{6}R$ (D) $\sqrt{3}\,R$

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Multiple Correct Answer Type 6 Q. [4 M (–1)]

9. A conducting ring is placed in a uniform external magnetic field present in the space within the ring and perpendicular to plane of ring. The magnetic field changes at a constant rate due to which a current of magnitude 4 amp flows in the ring. The resistance of parts PQR and PSR are 4Ω and 8Ω respectively. Then :-



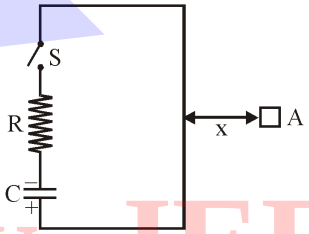
(A) The emf induced in the ring 48 V

(B) Potential difference between P & R is 24 V

(C) Potential difference between P & R is 8V

(D) Potential difference between P & R is zero as conservative electric field in the ring wire is zero

10. In figure the switch is closed at t = 0, with the capacitor of capacity 1 μF and having initial charge of 20 μC (the polarity shown). The left square loop has very large dimensions as compared to the distance between loops and a resistance of 10 Ω is connected in series with capacitor as shown. A wire of length 8 mm and resistance per unit length of 0.5 Ω/mm is bent in the form of square loop A and placed at a distance x = 1 m from left loop.



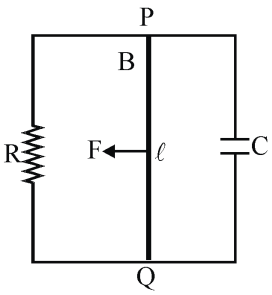
(A) The mutual inductance of the system will be 8×10^{-13} H

(B) The induced current in the loop A will be clockwise

(C) The induced current in the loop A will be anticlockwise

(D) The induced current in the loop A at time t = 2RCℓn2 will be 10^{-8} A

11. A conducting rod PQ of length ℓ is dogged with a constant force F along two smooth parallel rails separated by a distance ℓ as shown in the adjacent figure. Then choose the correct statement(s).



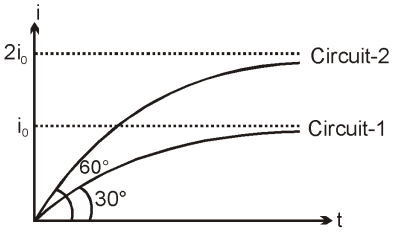
(A) terminal velocity of the rod, $v_t = \dfrac{2FR}{B^2\ell^2}$

(B) terminal velocity of the rod, $v_t = \dfrac{FR}{B^2\ell^2}$

(C) maximum charge on the capacitor, $q_{max} = \dfrac{FCR}{B\ell}$

(D) maximum charge on the capacitor, $q_{max} = \dfrac{2FCR}{B\ell}$

12. Growth of current in two different L-R circuits are depicted by the i-t graphs shown. Angle subtended by the curves with time axis at time t = 0 are also shown in the graph. τ_1 and τ_2 are time constants for the circuits 1 and 2 respectively. Choose the correct alternative



(A) $\dfrac{\tau_1}{\tau_2} = \dfrac{2}{3}$

(B) $\dfrac{\tau_1}{\tau_2} = \dfrac{3}{2}$

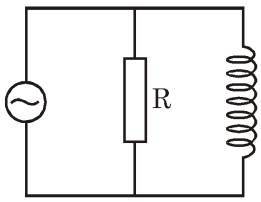
(C) Initial rate of change of current for circuit-2 is 2 times that of circuit-1

(D) Initial rate of change of current for circuit-2 is 3 times that of circuit-1

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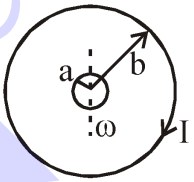
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13. In a circuit, alternating voltage with amplitude U_o = 100 V. is applied. The phase difference between current and voltage through the source is equal $\varphi = \pi/4$. R = 100 Ohm and the inductance is resistance free. Take ω = 100 rad/s



(A) The amplitude of current is $\sqrt{2}$ amp.
(B) The average power dissipated is 50 W
(C) The inductance is 1H
(D) The amplitude of current through the inductance is 0.5 A

14. A very small circular loop of radius a is initially coplanar and concentric with a much larger circular loop of radius b (>>a). A constant current I is passed in the large loop which is kept fixed in space and the small loop is rotated with angular velocity ω about a diameter. The resistance of the small loop is R & its self inductance is negligible. The current in the larger loop is clockwise. (Here θ is to be measured from axis of large ring)



(A) The current in the small loop as a function of time is $\frac{\pi a^2 \mu_0 I \omega \sin \omega t}{2bR}$

(B) The current in the small loop as a function of time is $\frac{\pi a^2 \mu_0 I \omega \cos \omega t}{2bR}$

(C) Torque that must be exerted on the small loop to rotate it, is $\frac{\omega}{R}\left(\frac{\pi b^2 \mu_0 I \sin \omega t}{2b}\right)^2$

(D) Torque that must be exerted on the small loop to rotate it, is $\frac{\omega}{R}\left(\frac{\pi a^2 \mu_0 I \sin \omega t}{2b}\right)^2$

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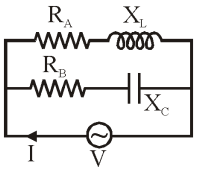
Linked Comprehension Type (2 Para × 2Q.) (2 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question No. 15 and 16

Figure shows a two branched parallel circuit with

$R_A = 20\Omega$, $X_L = 15\ \Omega$, $R_B = 6\Omega$ and $X_C = 8\Omega$.

The total current supplied by alternating source V is $I = \sqrt{29}$ A.

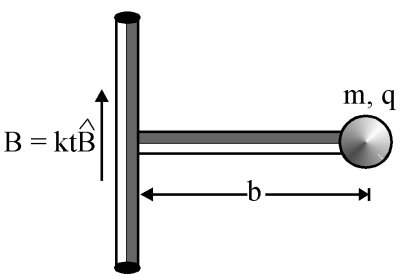


15. Current passing through the inductor (X_L) is :-
(A) Zero (B) 4A (C) 2 A (D) 5 A

16. Power dissipated in the circuit is :-
(A) Zero (B) 380 W (C) 920 W (D) 1400 W

Paragraph for Question Nos. 17 & 18

In the figure a small dielectric body of mass m carries charge q. The body lies at the end of a rod of length b, having negligible mass. The other end of the rod is connected to a vertical shaft mounted on frictonless bearings so that the rod is perpendicular to the shaft and thus free to swing in a horizontal plane. At time t = 0, the rod is at rest, and a uniform magnetic field is turned on whose magnitude increases linearly with time : B = kt, where k is a constant. The direction of the field is parallel to the shaft.



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17. What is the angular acceleration of the body about the shaft.

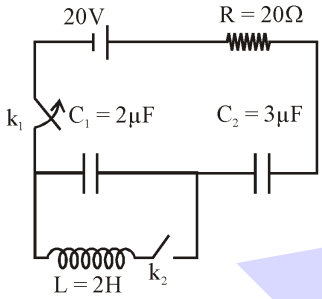
(A) $\alpha = \frac{qk}{2m}$ (B) $\alpha = \frac{2qk}{m}$ (C) $\alpha = \frac{qk}{m}$ (D) $\alpha = \frac{2qk}{3m}$

18. What is the speed v of the body as a function of time ?

(A) $v = \frac{2qkbt}{2m}$ (B) $v = \frac{qkbt}{2m}$ (C) $v = \frac{2qkbt}{m}$ (D) $v = \frac{2qkbt}{3m}$

Paragraph for Question Nos. 19 to 21

A circuit shown in the figure in which k_1 is closed and k_2 is open. Inductor L can be connected to capacitor C_1 by closing switch k_2 and opening k_1.



19. Let switch k_1 is closed and k_2 is opened for long time. The charge on capacitor C_2 will be :-

(A) 60 μC (B) 12 μC (C) 24 μC (D) 8 μC

20. At t = 0, when capacitors are fully charged, switch k_1 is opened and switch k_2 is closed, so that inductor is connected in series with capacitor C_1. The maximum charge will appear on capacitor C_1 for first time after time t = 0 is :-

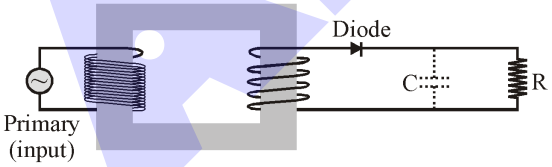
(A) $\frac{\pi}{2000}$ sec (B) $\frac{\pi}{1000}$ sec (C) $\frac{\pi}{500}$ sec (D) $\frac{\pi}{250}$ sec

21. The maximum energy across inductor will be :-

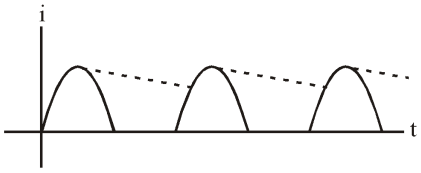
(A) 0.144 mJ (B) 0.288 mJ (C) 0.072 mJ (D) None of these

Paragraph for Question Nos. 22 to 24

Portable electronic devices such as radios and compact disc players are often powered by direct current supplied by batteries. Many devices come with AC-DC converters such as that shown in figure (i). Such a converter contains a transformer that steps the voltage down from 120 V to, typically , 9V and a circuit that converts alternating current to direct current. The AC-DC converting process is called rectification, and the converting device is called a rectifier.

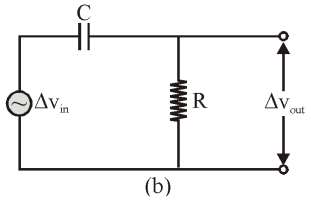


(i) A half-wave rectifier with an optional filter capacitor



(ii) Current versus time in the resistor. The solid curve represents the current with no filter capacitor, and the dashed curve is the current when the circuit includes the capacitor.

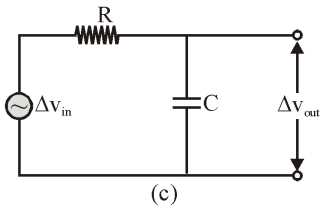
We can also design filters that respond differently to different frequencies. Consider the simple series RC circuit shown in figure. The input voltage is across the series combination of the two elements. The output is the voltage across the resistor.



(b)

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Now consider the circuit shown in figure, where we have interchanged the resistor and capacitor and where the output voltage is taken across the capacitor. At low frequencies, the reactance of the capacitor and the voltage across the capacitor is high. As the frequency increases, the voltage across the capacitor drops. Therefore, this filter is an RC low-pass filter.



(c)

22. Consider the filter circuit shown in Figure (b), the ratio of the input, output voltage is :-

(A) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{R}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

(B) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{R}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

(C) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{R}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

(D) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{R}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

23. What value of $\frac{\Delta v_{out}}{\Delta v_{in}}$ does approach as the frequency decreases toward zero ?

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) Zero

24. Consider the filter circuit shown in figure (c). The ratio of the input, output voltage is :-

(A) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{1/\omega C}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

(B) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{1/\omega C}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

(C) $\frac{\Delta v_{out}}{\Delta v_{in}} = \frac{1/\omega C}{\sqrt{R^2 - \left(\frac{1}{\omega C}\right)^2}}$

(D) $\frac{\Delta v_{in}}{\Delta v_{out}} = \frac{1/\omega C}{\sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}}$

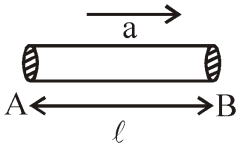
Linked Comprehension Type (1 Para × 2 Q.) [4 M (–1)]
(Multiple Correct Answer Type)

Paragraph for Questions 25 & 26

If a conductor is accelerated, initially the free electrons lag behind, thus setting up an electric field in the conductor. In steady state this field exerts force on free electrons and make them accelerate along with the conductor. (Historically this was used by Tolman-Stewant in 1916 for their experiments to measure charge to mass ratio).

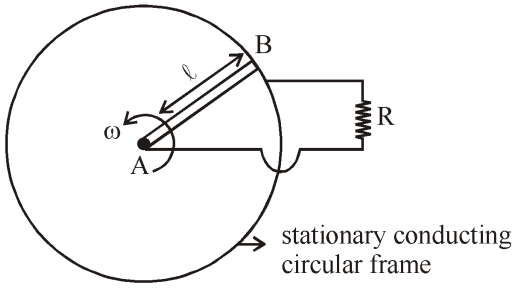
25. If the above rod accelerates with acceleration $\vec{a}$ along rod length and v_A and v_B are potential of end A and B respectively, then

(A) Potential of point A is less than that of B
(B) Potential of point B is less than that of A
(C) A constant electric field appears in the rod from B to A
(D) A constant electric field appears in the rod from A to B



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26. Figure shows a rod AB that can rotate with uniform angular velocity ω over a stationary conducting circular frame and resistance R is connected between centre A and frame.



(A) If the current was found to be I then the charge to mass ratio of electrons would be $\frac{\omega^2\ell^2}{2IR}$

(B) If the current was found to be I then the charge to mass ratio of electrons would be $\frac{2\omega^2\ell^2}{IR}$

(C) If a magnetic field B is applied perpendicular to plane of the paper then induce current (due to motional emf only) in rod will be $\frac{1}{2}\frac{B\ell^2\omega}{R}$

(D) If a magnetic field B is applied perpendicular to plane of the paper then induce current in rod will be radially outward

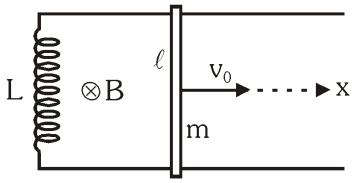
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SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 3 Q. [4 M (0)]

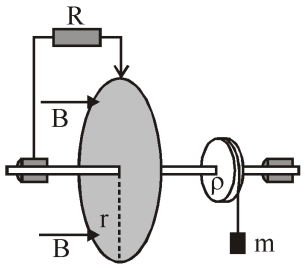
27. Two coaxial long solenoids of equal lengths have current i_1, i_2, number of turns per unit length n_1, n_2 and radius 3m, 5m respectively. If $n_1i_1 = n_2i_2 = 250$ (in SI unit) and the two solenoids carry current in opposite sense. Find the magnetic energy stored per unit length (in SI unit). (Take $\pi^2 = 10$)

28. A loop is formed by two parallel conductors connected by a solenoid with inductance L = 2H and a conducting rod of mass m = 80 g, which can freely (without friction) slide over the conductors. The conductors are located in a horizontal plane in a uniform vertical magnetic field with induction B = 4T. The distance between the conductors is equal to ℓ = 4cm. At the moment t = 0 the rod is imparted an initial velocity v_0 = 2 cm/s directed to the right. Find the amplitude (in cm) for the oscillations of conducting rod, if the electric resistance of the loop is negligible.



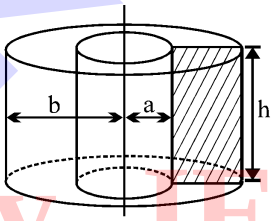
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29. A disc of radius r is made of a material of negligible resistance and can rotate about a horizontal shaft. A smaller disc of radius ρ is fixed onto the same shaft and has a massless cord wrapped around it, which is attached to a small object of mass m as shown. Two ends of a resistor of resistance R are connected to the perimeter of the disc and to the shaft by wiping contacts. The system is then placed into a uniform horizontal magnetic field B and mass m is released. Find the constant angular velocity (in rad/s) with which the disc will rotate after a certain time. If your answer is ω give value of $\frac{\omega}{20}$. Data : r = 10 cm, ρ = 2 cm, R = 0.01 Ω, B = 0.2 T, m = 50 g.



Subjective Type **1 Q. [4 M (0)]**

30. A ring of a rectangular cross section (see figure) is made of a material whose resistivity is ρ. The ring is placed in a homogeneous magnetic field. The intensity of the magnetic field is directed along the axis of the ring and increases directly with times, B = kt. Find the current induced in the ring.



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GR # EMI & AC **ANSWER KEY**

SECTION-I

Single Correct Answer Type **8 Q. [3 M (–1)]**

1. Ans. (C) **2. Ans. (B)** **3. Ans. (C)** **4. Ans. (B)** **5. Ans. (B)**
6. Ans. (D) **7. Ans. (A)** **8. Ans. (B)**

Multiple Correct Answer Type **6 Q. [4 M (–1)]**

9. Ans. (A,C) **10. Ans. (A,B,D)** **11. Ans. (BC)** **12. Ans. (B,D)** **13. Ans. (A,B,C)**
14. Ans. (A,D)

Linked Comprehension Type (2 Para × 2Q.) (2 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

15. Ans. (C) **16. Ans. (B)** **17. Ans. (A)** **18. Ans. (B)** **19. Ans. (C)**
20. Ans. (C) **21. Ans. (A)** **22. Ans. (C)** **23. Ans. (D)** **24. Ans. (A)**

Linked Comprehension Type (1 Para × 2 Q.) [4 M (–1)]
(Multiple Correct Answer Type)

25. Ans. (A, C) **26. Ans. (A,C)**

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

27. Ans. 2 **28. Ans. 5** **29. Ans. 5**

Subjective Type **1 Q. [4 M (0)]**

30. Ans. $\frac{k\,h}{4\rho}(b^2 - a^2)$

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